Waste Segregation Arduino Code

```
#include <ESP32Servo.h>
#include <Stepper.h>

// Sensors
#define RAIN_SENSOR 13
#define IR_SENSOR 15
#define PROX_SENSOR 2

// Servo
#define SERVO_PIN 14

// Stepper setup
const int stepsPerRevolution = 2048;
Stepper stepperMotor(stepsPerRevolution, 4, 12, 16, 1);

Servo gateServo;

// Bin positions
int currentPosition = 0;
int dryBin = 0;
int wetBin = 700;
int metalBin = 1400;

void setup() {
  Serial.begin(115200);

  pinMode(RAIN_SENSOR, INPUT);
  pinMode(IR_SENSOR, INPUT);
  pinMode(PROX_SENSOR, INPUT);

  gateServo.setPeriodHertz(50);
  gateServo.attach(SERVO_PIN, 500, 2400);
  gateServo.write(0);

  stepperMotor.setSpeed(10);

  Serial.println("Waste Segregation System Started");
}

void loop() {
  int dryWaste = digitalRead(RAIN_SENSOR);
  int wetWaste = digitalRead(IR_SENSOR);
  int metalWaste = digitalRead(PROX_SENSOR);

  if (metalWaste == HIGH) {
    Serial.println("Metal Waste Detected");
    moveToBin(metalBin);
    openGate();
  }
  else if (dryWaste == HIGH) {
    Serial.println("Dry Waste Detected");
    moveToBin(dryBin);
    openGate();
  }
  else if (wetWaste == HIGH) {
    Serial.println("Wet Waste Detected");
    moveToBin(wetBin);
    openGate();
  }

  delay(500);
}

void moveToBin(int targetPosition) {
  int stepsToMove = targetPosition - currentPosition;
  stepperMotor.step(stepsToMove);
  currentPosition = targetPosition;
  delay(500);
}

void openGate() {
  gateServo.write(90);
  delay(1500);
```

```
  gateServo.write(0);
  delay(1000);
}
```